

Parental Imprinting of the Mouse Insulin-like Growth Factor II Gene

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Summary

We are studying mice that carry a targeted disruption of the gene encoding insulin-like growth factor II (IGF-II). Transmission of this mutation through the male germline results in heterozygous progeny that are growth deficient. In contrast, when the disrupted gene is transmitted maternally, the heterozygous offspring are phenotypically normal. Therefore, the difference in growth phenotypes depends on the type of gamete contributing the mutated allele. Homozygous mutants are indistinguishable in appearance from growth-deficient heterozygous siblings. Nuclease protection and in situ hybridization analyses of the transcripts from the wild-type and mutated alleles indicate that only the paternal allele is expressed in embryos, while the maternal allele is silent. An exception is the choroid plexus and leptomeninges, where both alleles are transcriptionally active. These results demonstrate that IGF-II is indispensable for normal embryonic growth and that the IGF-II gene is subject to tissue-specific parental imprinting.

Introduction

Genes encoding growth factors and their receptors are candidates for key regulatory elements that could mediate decisions in developmental pathways by providing products necessary for cell growth and/or differentiation during embryogenesis (see Whitman and Melton, 1989, for a recent discussion). The involvement of members of growth factor families in the process of mesoderm induction in amphibia has provided a paradigm consistent with this view (reviewed by Slack et al., 1989; Smith et al., 1989; Wylie, 1990; Slack, 1991). Moreover, it was shown that the mouse mutations *W* and *Sl*, both of which affect the formation of a number of embryonic cell lineages, lie in the genes encoding the *c-kit* receptor and its cognate ligand MGF (mast cell growth factor), respectively (reviewed by Pawson and Bernstein, 1990; Witte, 1990). Recently, we provided direct evidence for a physiological role of the insulin-like growth factor II in embryonic growth (DeChiara et al., 1990).

The insulin-like growth factors I and II (IGF-I and IGF-II; recently reviewed by Daughaday and Rotwein, 1989; Humbel, 1990), which have a structural similarity to proinsulin, are mitogenic polypeptides for a variety of cell lines. Postnatally, IGF-I mediates many of the growth-promoting effects of growth hormone by binding to the IGF-I (type 1) receptor, a ligand-activated tyrosine kinase. The type 1

receptor recognizes, albeit less avidly, IGF-II and also cross-reacts weakly with insulin (reviewed by Czech, 1989). Another structurally unrelated receptor (type 2), which might not actively participate in signal transduction, binds IGF-II with high affinity, cross-reacts very weakly with IGF-I, and does not recognize insulin (reviewed by Roth, 1988; Czech, 1989; see also Discussion).

The production of IGFs in many tissues suggested that they play autocrine/paracrine roles. However, an additional endocrine function cannot be excluded because, unlike most other growth factors, the IGFs circulate in the plasma associated with IGF-binding proteins (reviewed by Baxter and Martin, 1989). The concentrations of IGF-II in fetal plasma and tissues are high, but decline progressively postnatally. This led to an early suggestion about a possible involvement of IGF-II in embryonic processes (Moses et al., 1980; D'Ercole et al., 1980). The significance of growth factors in embryogenesis was also implied from observations suggesting that endocrine gland products, for example growth hormone, are not involved in prenatal development (reviewed by Milner and Hill, 1987; Chard, 1989; Girard, 1989).

In both rodents and humans, the IGF-II gene is single copy and linked to the insulin gene locus in the same transcriptional orientation (Soares et al., 1986; Bell et al., 1985). These genes reside in syntenic regions of rat chromosome 1 (Soares et al., 1985, 1986), mouse chromosome 7 (Lalley and Chirgwin, 1984; Ken Rosen, personal communication), and human chromosome 11 (Harper et al., 1981; Brissenden et al., 1984; Tricoli et al., 1984; de Pagter-Holthuis et al., 1985).

The rat IGF-II gene, which has been extensively characterized (Soares et al., 1986; Frunzio et al., 1986; Ueno et al., 1987; Evans et al., 1988; Ikejiri et al., 1990; reviewed by Sussenbach, 1989; Gammeltoft, 1989), utilizes three promoters (P0, P1, and P2), without apparent tissue specificity, which generate alternate 5' noncoding exons (E1.0, E1.1, and E1.2). The two predominant mature mRNAs (4.5 and 3.5 kb) consist of either E1.1 or E1.2, respectively, connected to three additional exons (E2, E3, and E4). P0 and multiple polyadenylation signals in E4 (Ueno et al., 1989) contribute to the appearance of other, less abundant transcripts. Despite this complexity, the IGF-II gene encodes a single known polypeptide. The organization of the mouse IGF-II gene is very similar to that of the rat homolog.

In situ hybridization analysis of the developmental pattern of IGF-II gene expression in rodents has demonstrated that transcripts appear after implantation, first in all of the trophectodermal derivatives, and then in primitive endoderm, extraembryonic and embryonic mesoderm, and the lining of the foregut (Lee et al., 1990). Thus, the IGF-II gene is expressed in all of the components forming the chorioallantoic placenta and in specific cell lineages of the embryo proper. In embryos of more advanced stages, the IGF-II gene is active in most tissues derived

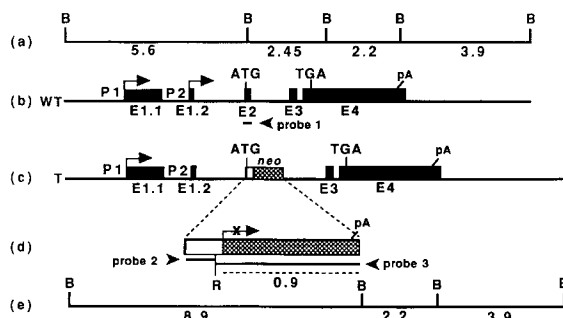


Figure 1. Restriction Maps of the WT and Targeted Mouse IGF-II Alleles

(a and e) The BamHI (B) sites and the distances (in kb) between these sites are shown in the maps of the WT and targeted (T) genes, respectively. (b) The positions of five exons (E1.1–E4; black boxes) of the WT IGF-II gene relative to the map in (a) are indicated. P1 and P2 are promoters from which the alternate 5' noncoding exons E1.1 and E1.2 are initiated (arrows). The positions of the initiation codon (ATG), termination codon (TGA), and major polyadenylation signal (pA) are shown. The region of the E2 exon represented in the E2 riboprobe (probe 1; see Experimental Procedures) is indicated. (c) In the targeted gene, in which only the IGF-II P1 promoter seems to be active (see text), a *neo* cassette (1.15 kb) has replaced exon E2 and about 0.1 kb of the downstream intron (total of 0.25 kb). (d) Enlargement of the cassette segment, consisting of the coding region of the *neo* gene (stippled box) and regulatory sequences (open box). The latter are a mutant polyoma enhancer and the herpes simplex virus *tk* promoter that are apparently unable to initiate transcription (crossed-out arrow) of a ~0.9 kb RNA when present in the mutated gene (see text). Positions of two *neo* riboprobes (probe 2, used in nuclease protection experiments; probe 3, used for in situ hybridization; see Experimental Procedures) are shown.

from somites, lateral plate, and cardiac mesoderm, while expression in derivatives of the other germ layers is restricted to a few sites (Stylianopoulou et al., 1988a). Transcripts are not detected in the ectoderm-derived central and peripheral nervous system at any embryonic stage (Lee et al., 1990; Stylianopoulou et al., 1988a). The transcriptional activity of the IGF-II gene continues during the early neonatal period and then ceases in most tissues. In adult rodents, expression of the IGF-II gene is maintained only in the choroid plexus and the leptomeninges (Stylianopoulou et al., 1988b).

The pattern of expression of the IGF-II gene during rodent embryogenesis, in addition to the other evidence summarized above, suggested that the cognate growth factor has a developmental role. To examine this role directly, we have disrupted one of the IGF-II alleles in cultured mouse embryonic stem (ES) cells by gene targeting (DeChiara et al., 1990). Unexpectedly, germline transmission of the mutated gene from male chimeras resulted in growth-deficient heterozygous progeny whose weight was approximately 60% of that of ES cell-derived wild-type littermates. Here we present genetic evidence indicating that manifestation of the mutant phenotype depends on the sex of the parent contributing the disrupted allele. These data, in conjunction with the results of a molecular analysis, demonstrate that the endogenous IGF-II locus is subject to parental imprinting.

Results

An Animal Model to Study the In Vivo Role of IGF-II

We have generated a novel strain of transgenic mice carrying a defined deletion at the endogenous IGF-II gene locus by targeted mutagenesis (DeChiara et al., 1990). In the targeted IGF-II allele, a DNA cassette (1.15 kb; Thomas and Capecchi, 1987) containing the bacterial neomycin resistance gene (*neo*) replaced 0.25 kb of genomic sequence that includes the first coding exon (E2) of the IGF-II gene (Figure 1). Thus, the disrupted gene cannot generate a functional polypeptide because the deleted exon encodes the signal peptide (24 amino acids) and 28 of the 67 residues of mature IGF-II, including the putative receptor binding site (Humbel, 1990). This mutation was transmitted through the germline of chimeric male animals constructed from a targeted ES cell clone. In the heterozygous progeny of these chimeras, the wild-type (WT) and mutated alleles and their transcripts can be readily discriminated using specific E2 and *neo* sequence probes, respectively.

When the disrupted IGF-II gene was transmitted from founder male chimeras, all progeny that inherited the mutation exhibited a growth deficiency phenotype (DeChiara et al., 1990). This correlation between heterozygosity and growth deficiency at birth was observed in many subsequent litters obtained from crossing male chimeras to WT females of various genetic backgrounds (Table 1A and Figure 2). We call these small heterozygous progeny "paternal heterozygotes" (P-heterozygotes) to distinguish them from heterozygous offspring that inherit the mutated allele maternally (M-heterozygotes).

The P-heterozygotes were viable and fertile. They were normally proportioned miniature animals (proportionate dwarfs) and exhibited the same rates of weight gain as their WT littermates. Thus, the mutant to normal body weight ratio (60%) remained constant throughout adult life. The growth deficiency phenotype was also evident in both the fetal and placental portions of P-heterozygous conceptuses, examined only from embryonic day 16 (e16) onward. From these observations we concluded that the mutation was exerting an effect on the overall growth of the conceptus in utero.

The IGF-II Gene Is Subject to Parental Imprinting: Genetic Evidence

To examine the genetic basis of the unexpected appearance of growth deficiency in heterozygotes more rigorously, we initiated a breeding program by mating P-heterozygous F1 progeny of both sexes to WT partners. The resulting F2 progeny were scored phenotypically at birth, marked, and subsequently genotyped. We observed that, when the mutated allele was paternally inherited, all litters of F2 progeny consisted of WT animals and dwarf heterozygotes in approximately equal ratios. In contrast, all of the F2 progeny derived from P-heterozygous females were phenotypically identical and exhibited normal body size, although genotyping showed that approximately half of them were heterozygous (Table 1A). Dwarf (P) and

Table 1. Breeding Data

			Number of Progeny ^a					
Parents		Generation	Normal		Growth Deficient		Number of Litters	
Male	Female		WT	M-Het	P-Het	Hom		
A. Outcrosses								
Chimera	x WT	F1 ^b	43		42		12	
Het	x WT	F2	27		28		7	
Het	x WT	F3	27		29		7	
Het	x WT	F4	13		6		2	
WT	x Het	F2	17	20			5	
WT	x Het	F3	26	24			7	
WT	x Het	F4	0	5			1	
B. Intercrosses								
Het	x Het	F2 ^c	51	45	49	34	24	

^a In the column headings, the phenotype is listed above the genotypes.

^b Only 25 normal and 16 growth-deficient animals of the F1 were genotyped and found to be WT and heterozygous (Het), respectively. In the other generations, all animals were genotyped.

^c From additional intercrosses (six litters), only the growth-deficient progeny (29) were studied. Genotyping showed that 13 animals were heterozygous and 16 animals were homozygous (Hom).

normal-sized (M) F2 heterozygotes of both sexes were then mated to WT partners. Again, the growth deficiency was manifested in heterozygous F3 progeny only when the mutated gene was paternally derived. The analysis was extended to the F4 generation with the same results.

One of the pedigrees demonstrating this phenomenon is shown in Figure 2.

Since all of the heterozygotes possess the mutated IGF-II allele, the phenotypic difference in size between the P- and M-heterozygotes can be attributed to differential expression of the intact allele, which is dependent on the gamete of origin. Therefore, the intact paternal allele in M-heterozygotes is presumably transcriptionally active and these animals are normal in size. In contrast, the intact maternal allele in P-heterozygotes is presumably underexpressed or even silent, since these animals are always growth deficient. The implication of this genetic evidence is that the IGF-II locus is subject to parental imprinting (see Discussion).

However, to invoke imprinting, another criterion should be met: It must be shown that the function of an IGF-II allele is reversibly switched by passage through the germline of opposite sex in successive generations. This was demonstrated by mating P- or M-heterozygotes to genotypically WT siblings and obtaining results identical to those described above (Figure 2). In these matings the switch of the intact allele from an expressing to a nonexpressing state (or vice versa) could be deduced from the phenotypes of the progeny.

Homozygous and Heterozygous IGF-II Mutant Mice Are Phenotypically Indistinguishable

Further genetic evidence in support of parental imprinting was provided from the results of intercrosses between P-heterozygous IGF-II mutant animals, performed to study the consequences of the mutation in the homozygous

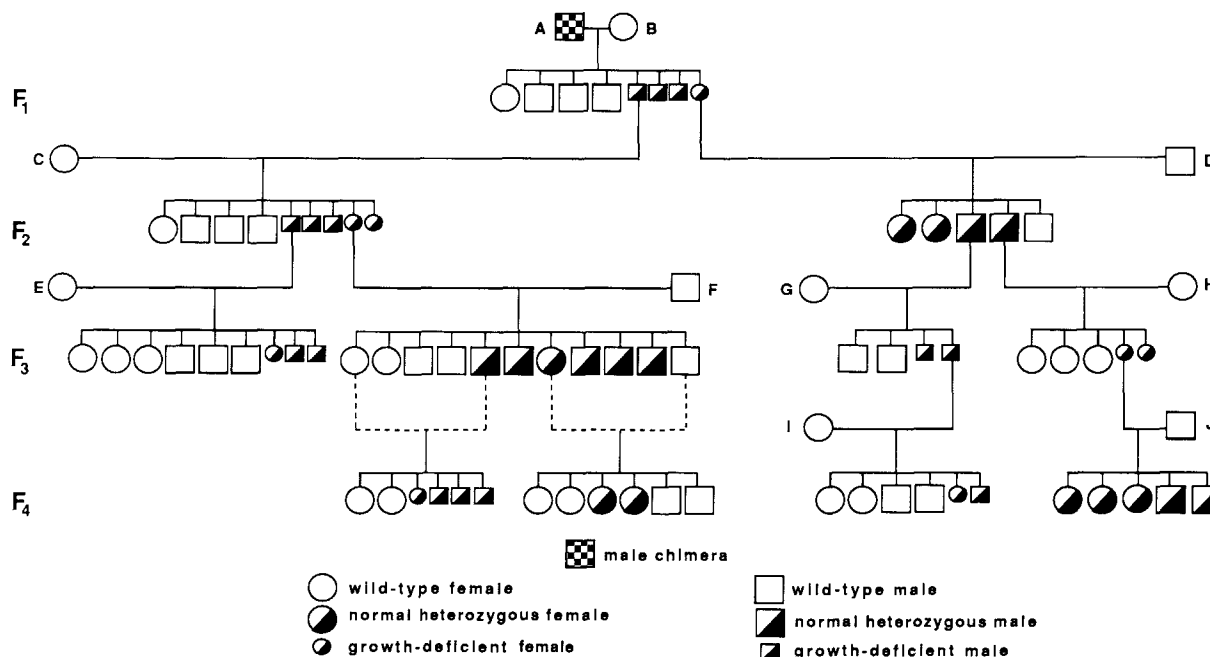


Figure 2. Partial Pedigree of IGF-II Mutant Animals

Four generations of offspring from a complete germline male chimera (animal A; DeChiara et al., 1990) are shown in a partial pedigree. The genetic backgrounds of the animals used in these crosses were: A, C, D, G, H, I, and J, 129/Sv/Ev; B, (C57BL/6J x DBA/2)F1; C, MF1; E and F, C57BL/6J. Solid and dashed lines represent outcrosses and intercrosses, respectively.

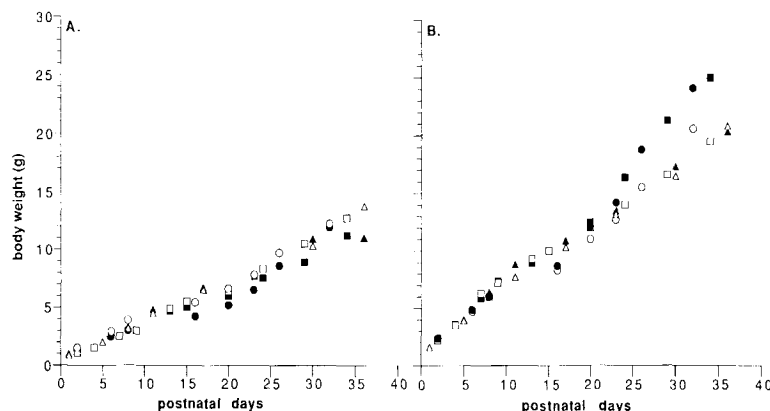


Figure 3. Growth Curves of Progeny Derived from Heterozygous Intercrosses

(A) Weights of growth-deficient heterozygous (open symbols) and homozygous (filled symbols) progeny from three litters are plotted. The numbers of heterozygotes and homozygotes, respectively, were: litter 1 (circles), 3 and 1; litter 2 (squares), 1 and 2; and litter 3 (triangles), 5 and 1. Weights of animals of each category, which were not significantly different, were averaged. The average of heterozygous to homozygous weight ratios for all time points was 1.2 ± 0.09 (litter 1), 1.06 ± 0.06 (litter 2), and 1 ± 0.13 (litter 3).

(B) Weights of WT animals (open symbols) and normal-sized heterozygotes (filled symbols) from the same three litters as in (A) are plotted. The numbers of WT animals and heterozy-

gotes, respectively, were: litter 1, 3 and 1; litter 2, 2 and 2; and litter 3, 1 and 1. Weights were averaged as in (A). The average of WT to heterozygous weight ratios was 0.92 ± 0.06 (litter 1), 0.96 ± 0.15 (litter 2), and 0.96 ± 0.04 (litter 3). The equivalent ratio in a litter derived from an outcross (4 WT animals and 2 M-heterozygotes; not shown) was 1 ± 0.02 .

The percentage increase in weight at each time point, as compared with the weight at birth, was calculated for all progeny of the intercrosses. The ratios of such values between any two animals, regardless of phenotype, did not deviate significantly from unity.

state. Genotyping indicated that three classes of progeny were recovered from the intercrosses at frequencies conforming to the predicted Mendelian ratios (WT, heterozygous, and homozygous progeny in a 1:2:1 ratio; Table 1B). Phenotypically, however, the offspring were either normal or growth deficient in a 1:1 ratio, exhibiting practically uniform body weight within each class. Thus, considering both the genotype and the phenotype, there were four categories of progeny: WT animals with normal weight, heterozygotes with normal weight (presumably M-heterozygotes), growth-deficient heterozygotes (presumably P-heterozygotes), and growth-deficient homozygotes.

Follow-up comparisons of homozygous mutant offspring and P-heterozygous siblings indicated that the two categories of mutant animals exhibit similar viability and fertility and practically identical postnatal growth rates (Figure 3A). Overall, these two categories of dwarf mice are phenotypically indistinguishable, although we cannot exclude the possibility that some subtle differences exist. A similar comparison between WT animals and M-heterozygotes, including an assessment of postnatal growth characteristics (Figure 3B), also indicated complete similarity at the phenotypic level. It is notable that there is no difference in the rate of weight gain as a function of weight at birth between the two phenotypically distinct classes of offspring (see legend to Figure 3). This is further evidence indicating that the main function of the IGF-II gene is indeed exerted during the embryonic period.

The results obtained from intercrossing P-heterozygotes, in conjunction with the other genetic evidence, support our interpretation that the IGF-II locus is subject to parental imprinting. The homozygous mutant mice, in which both alleles are disrupted, do not exhibit any apparent features suggestive of a more severe phenotype than that of P-heterozygous dwarfs. On the other hand, the M-heterozygotes, which possess only a single intact allele, are phenotypically normal. From these data we can conclude that, as a consequence of parental imprinting, only the

paternally derived IGF-II allele is expressible. A corollary is that, at least in the case of the IGF-II gene, the effect of the imprinting process on functionality can be considered practically complete because no intermediate phenotypes have been recovered.

The IGF-II Gene Is Subject to Parental Imprinting: Molecular Evidence

Preliminary quantitations of RNA by slot blot analyses indicated that the level of IGF-II transcripts from the intact maternal allele in P-heterozygous embryos did not exceed 10% of that detected in WT littermates (DeChiara et al., 1990), and it was probably much less.

To examine IGF-II gene expression more accurately, we used RNAase protection assays to analyze total cell RNA prepared from genotyped embryos (excluding the head region) and from their respective placentas. Each of the examined litters consisted of either P-heterozygous and WT littermates (P-group) or M-heterozygous and WT littermates (M-group). We examined three litters of the P-group, one at e16.5 and two at e18.5, and two litters of the M-group, one each at e13.5 and e18.5. RNA samples from respective placentas (e16.5 of the P-group and e13.5 of the M-group) were also examined. For the assays we used two riboprobes (see Figure 1 and Experimental Procedures), representing exon E2 and *neo* sequence, to determine the steady-state level of transcripts expressed from the intact and the mutated alleles, respectively. The particular *neo* probe that we selected could monitor transcripts initiated from an IGF-II promoter (see below). A cytoplasmic actin riboprobe was used as an internal control for RNA integrity.

Examples of these analyses are shown in Figure 4A. RNA from WT embryos (lane 1) or M-heterozygotes (lane 3) protected the E2 probe from nucleic acid digestion, while the *neo* probe was degraded (lanes 4 and 6, respectively). Exactly the opposite results were obtained with RNA from the P-heterozygous embryos, which was unable to protect

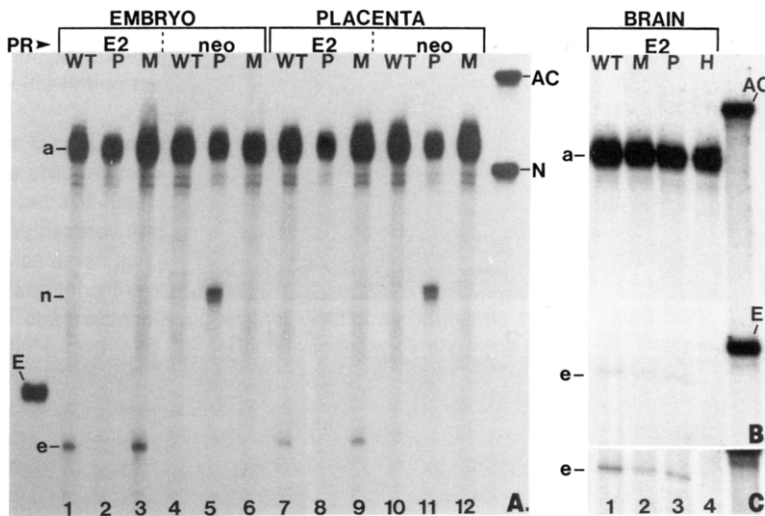


Figure 4. Nuclease Protection Analysis
(A) Samples of total cell RNA (20 μ g per lane) prepared from WT (ϵ 16.5), P-heterozygous (P; ϵ 16.5), and M-heterozygous (M; ϵ 13.5) embryos (excluding the head regions) and from their respective placentas were analyzed by RNAase protection, using actin (AC), *neo* (N), and exon E2 (E) riboprobes (PR). For the assays, the actin riboprobe was mixed with either the E2 or the *neo* probe. The corresponding protected fragments of the three probes are labeled a, n, and e (see Experimental Procedures). (B) Total brain RNA from 3-week-old WT, P-heterozygous, M-heterozygous, and homozygous (H) animals was analyzed by RNAase protection, using a combination of actin and E2 riboprobes, to assess IGF-II gene expression in the choroid plexus and leptomeninges. (C) is a longer exposure of the lower part of the autoradiogram in (B).

the E2 probe (Figure 4A, lane 2), while protection of the *neo* probe was observed (lane 5). Consistent data were obtained from all litters analyzed, and the results using RNA from each embryo and its cognate placenta were identical (Figure 4A, compare lanes 1–6 with lanes 7–12).

These molecular data fulfilled the predictions from the genetic evidence for imprinting, and demonstrated that the paternal IGF-II gene is transcriptionally active, whereas the maternal allele is not expressed at levels detectable by sensitive nuclease protection analysis. Thus if any transcription of the maternal allele occurs, it cannot exceed basal levels.

The transcripts containing *neo* sequence, which are generated from the paternally derived disrupted allele, have not been examined exhaustively. We note, however, that preliminary results from Northern and riboprobe protection analyses (data not shown) indicated that the exogenous regulatory sequences of the *neo* cassette (a mutant polyoma virus enhancer linked to the promoter of the herpes simplex virus *tk* gene) most likely do not contribute to transcription, when initiation events occur from an IGF-II promoter. This was inferred from the absence of an RNA species initiated at the *tk* promoter and terminating after the artificial polyadenylation signal of the *neo* cassette (expected size of about 0.9 kb; see Figure 1). Instead, three predominant RNA species of approximately 5.2, 3.5, and 2.5 kb were observed. All of these RNAs were initiated from the IGF-II P1 promoter and contained only the 5' portion of the *neo* coding region and transcribed upstream regulatory sequences, possibly as a result of aberrant splicing events. The mechanism by which the *tk* promoter–polyoma enhancer regulatory sequences and also the IGF-II P2 promoter are negatively regulated in the context of the mutated gene is unknown.

The Maternal IGF-II Allele Is Expressed Only in the Choroid Plexus and the Leptomeninges:

In Situ Hybridization Analysis

Imprinting might influence IGF-II gene expression in a stage-specific fashion during embryogenesis. For exam-

ple, our data did not exclude the possibility that the maternal allele is expressed transiently at embryonic stages prior to ϵ 13.5. Alternatively, despite the sensitivity of nuclease protection assays and the apparently complete effect of imprinting at the gross phenotypic level, maternal gene transcripts could be localized to one or a few restricted sites among the many embryonic tissue lineages that express the IGF-II gene (Lee et al., 1990). If this were the case, maternal transcripts present in total RNA might have remained undetectable, owing to dilution. To address these questions, we performed in situ hybridization analysis using sections of embryos recovered at ϵ 9.5 and ϵ 16.5. For these experiments, embryonic material was obtained from reciprocal matings between homozygous IGF-II mutant mice and WT partners. To compare in situ hybridization patterns accurately, we included sagittal sections from both P- and M-heterozygotes on the same slide, and used adjacent sections from each embryo for hybridization with either E2 or *neo* probes.

The results of this analysis (Figure 5) showed that in both embryonic stages examined, the transcripts from the paternal allele (monitored with a *neo* probe in P-heterozygotes and with an E2 probe in M-heterozygotes; see Figure 1) exhibited a pattern of localization in extraembryonic and embryonic tissues identical to that previously detected by in situ hybridization (Lee et al., 1990; Stylianiopoulou et al., 1988a). In contrast, transcripts from the maternal allele (monitored with an E2 probe in P-heterozygotes) were undetectable, with one very interesting exception. In sections of the ϵ 16.5 embryos, IGF-II gene transcripts localized to the choroid plexus and the leptomeninges were detected from both the paternal and maternal alleles (Figures 5c–5j). Since these structures, which are associated with the central nervous system, are the only sites of IGF-II gene expression in adult rodents, we examined whether the expression of both alleles persists postnatally. Nuclease protection analyses, using total brain RNA from 3-week-old animals, indicated that this is indeed the case (Figures 4B and 4C).

We note that when the *neo* probe was hybridized to sec-

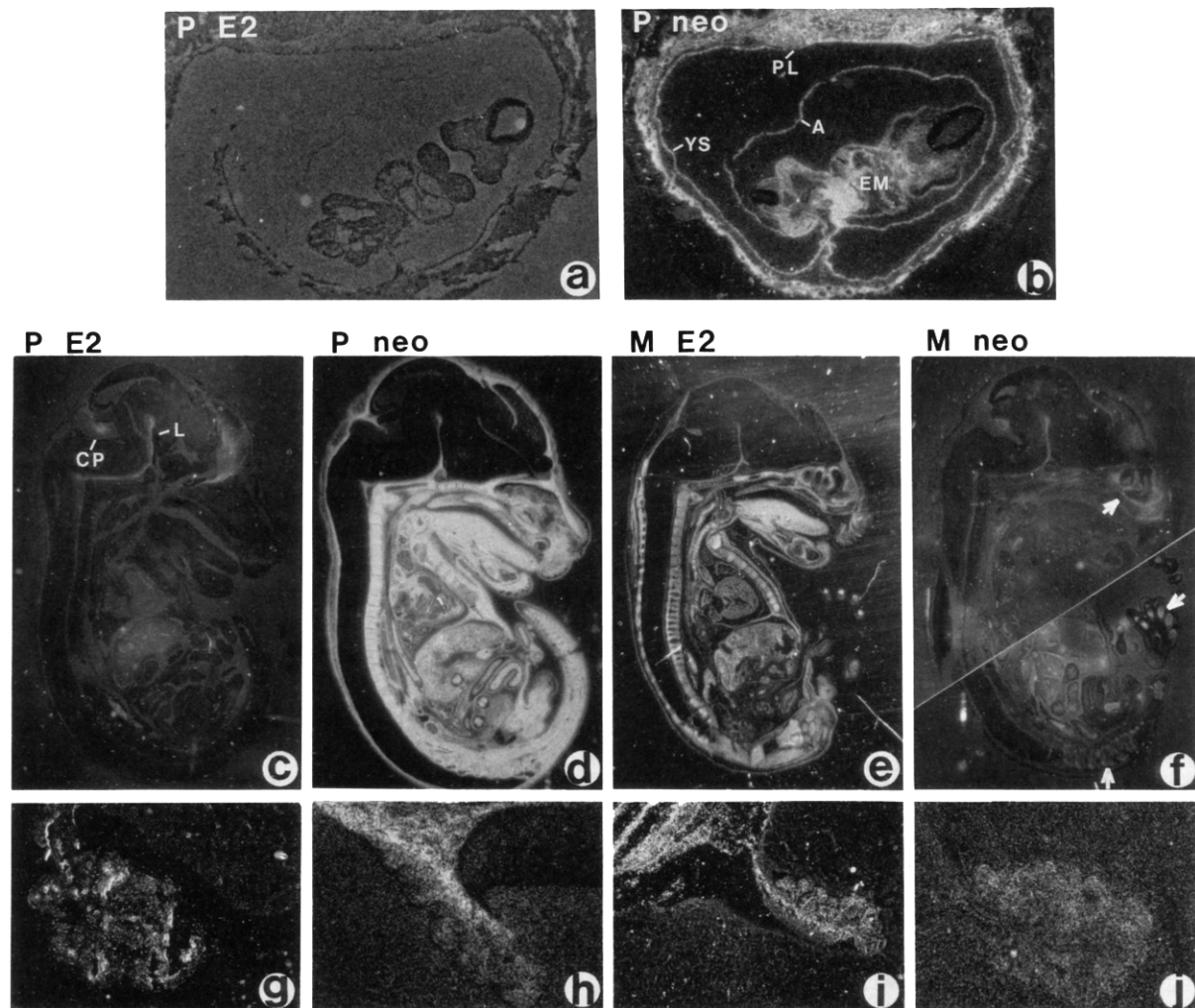


Figure 5. In Situ Hybridization Analysis

Sections of mouse embryos at e9.5 (a and b; transverse) and e16.5 (c–f; sagittal), either P-heterozygotes (P) or M-heterozygotes (M), were hybridized as indicated with E2 and *neo* probes (see Figure 1 and Experimental Procedures), processed, and photographed under dark-field illumination. In (b), hybridization was observed in the embryo proper (EM) and in extraembryonic structures (PL, developing placenta; A, amnion; YS, yolk sac). In all e16.5 embryos (c–f) hybridization was observed in the choroid plexus (CP) and the leptomeninges (L), regardless of probe. The corresponding plexi are presented at higher magnification (g–j) to show the localization of silver grains in this tissue. (f) is a composite of two different sections from the same embryo, for better display of cartilaginous structures (arrows) that exhibit a positive hybridization signal when examined at high power (not shown; see text). Regions of apparent signal in the abdominal regions of the embryos shown in (c) and (f) are photographic artifacts due to reflection (silver grains are absent from these regions). Hybridization with control sense-strand RNA did not give a signal above background (not shown).

tions of e16.5 M-heterozygous embryos, transcripts from the mutated maternal allele were detected not only in the plexus and the leptomeninges, but also in some cartilaginous structures present in the tail, the digits, and the region surrounding the eye (Figure 5f, arrows). However, when P-heterozygous embryos were examined with the E2 probe, expression of the intact maternal allele was not detected in these sites (Figure 5c). A possible explanation for this observation, which is currently under investigation, is that the viral regulatory sequences of the *neo* cas-

sette somehow function independently in these particular tissues.

Discussion

The major conclusion from our studies is that the IGF-II gene, which is involved in embryonic growth, is subject to parental imprinting. However, the choroid plexus and the leptomeninges (henceforth referred to as "plexus," for brevity) are excluded from the otherwise general influence

of imprinting on IGF-II gene expression. This novel observation places restrictions on possible models pertaining to the imprinting mechanism.

The Developmental Role of IGF-II

The proportionate dwarfism phenotype that we have observed in P-heterozygous and homozygous mutant mice demonstrates that the IGF-II polypeptide, apparently acting as a mitogen, plays an important developmental role determining the extent of embryonic growth. Moreover, our observations suggest that distinct mechanisms can affect growth and differentiation independently, because IGF-II, in its primary role, does not appear to act as a morphogen or an inductive signal (see Wolpert, 1989). The simplest view is that prior to e16, IGF-II is an indispensable participant in some crucial proliferation event(s). Subsequently, growth becomes IGF-II independent. However, this interpretation cannot account for the pattern of IGF-II gene expression during embryogenesis (Lee et al., 1990; Stylianopoulou et al., 1988a), which had implied a role(s) for the cognate growth factor in particular tissues and at certain developmental stages. Thus there may be overlap or compensation of function by other gene products affecting growth. We thought, for example, that IGF-I, which is also present in embryos (Bondy et al., 1990), could potentially play such a compensatory role. However, our preliminary analysis did not reveal any increase in IGF-I gene transcripts in late-gestation, P-heterozygous IGF-II mutant embryos (unpublished data).

Another question is whether the IGF-II dwarf phenotype is an indirect consequence of impaired placental trophic functions. Although not excluded, this interpretation is unlikely because it cannot easily account for the fact that the mutant to normal weight ratio (60%) is exactly the same for both the placenta and the embryo proper. A more straightforward explanation is that, in the absence of IGF-II, proliferation is impaired similarly in both the placenta and the embryo. Alternatively, the mutation exerts its effect before the allocation of extraembryonic and embryonic cell lineages. Although IGF-II gene expression was detected by *in situ* hybridization only after implantation (Lee et al., 1990), results of polymerase chain reaction assays have suggested that IGF-II transcripts might be present from the 2-cell stage onward (Rappolee et al., 1990).

Parental Imprinting in Mammals

In mice, and apparently in humans, the paternal and maternal members of some autosomal gene pairs are functionally nonequivalent. This differential behavior in gene expression that depends on parental legacy is attributed to parental imprinting (reviewed by Surani, 1986; Surani et al., 1986, 1990; Monk, 1988; Solter, 1988; Sapienza, 1989, 1990; Reik, 1989). In this regard, the IGF-II gene should be considered a specific example of a locus that receives a marking (imprint) introduced in the paternal or maternal germline during or before gametogenesis.

Previously, evidence that portions of the paternal and

maternal genomes play complementary roles has been provided by pronuclear transplantation experiments in mice, which demonstrated that isoparental embryos (either androgenetic or gynogenetic) do not develop to term (Barton et al., 1984; McGrath and Solter, 1984a; Surani et al., 1984; reviewed by Surani, 1986; Solter, 1988). This is corroborated by the observation in humans that only the paternal genome is present in complete hydatidiform moles, which are abnormal conceptuses consisting exclusively of trophoblastic elements (reviewed by Szulman and Surti, 1984).

Independent evidence showing functional nonequivalence of the parental genomes has been provided by genetic data. For example, the phenotypes of some human genetic disorders depend on paternal or maternal inheritance of the mutation (reviewed by Reik, 1989). In mice, extensive studies of the consequences of transmission of chromosomal rearrangements have demonstrated that combinations of maternal duplication/paternal deficiency (or vice versa) of certain chromosomes, or chromosome segments, either lead to lethality or produce opposite phenotypes depending on the particular disomy/nullisomy involved (reviewed by Searle and Beechey, 1978, 1985; Cattanaach, 1986; Searle et al., 1989; Beechey et al., 1990). For example, mice disomic for a paternally derived Robertsonian translocation of chromosome 11 were significantly larger than their littermates at birth. Conversely, they were smaller, if the disomy was derived maternally (Cattanaach and Kirk, 1985; Cattanaach, 1986). Presumably, the absence of an active paternal allele in the latter case resulted in a growth deficiency phenotype strikingly similar to that observed with the IGF-II mutation, both in terms of weight (60% of normal) and postnatal growth rates. Interestingly, the data from these disomics are compatible with an interpretation invoking the involvement of a single gene locus on chromosome 11; if an entire chromosomal segment were subject to imprinting, the absence of many paternal genes in the Robertsonian translocation would be lethal or at least would have additional phenotypic effects.

Whether the IGF-II gene also represents a solitary imprinted locus is unknown. We note, however, that it is highly unlikely that the neighboring insulin gene (gene-II) is subject to imprinting. This can be inferred from the fact that the genome of rodents contains a second unlinked insulin gene (gene-I) that is a functional retroposon (Soares et al., 1985), expressed at the same level as the ancestral gene-II. If gene-II were imprinted (for 1x adjustment of the functional gene dosage), it would have been impossible to explain the evolutionary fixation of an expressing retroposon by positive selection. Nevertheless, the possibility remains that the IGF-II gene might be located at one end of an imprinted gene cluster. Such a cluster could be part of a region of imprinted genes, inferred from genetic evidence, that lie in the distal segment of mouse chromosome 7 (Beechey et al., 1990). This segment may be syntenic with a region on the short arm of human chromosome 11 that contains the linked genes encoding tyrosine

hydroxylase–insulin–IGF-II (Glaser et al., 1989). In the mouse, the tyrosine hydroxylase gene, probably also linked to the insulin/IGF-II pair, has been mapped on distal chromosome 7 (Brilliant et al., 1987). On the other hand, independent imprinting of several genes residing in relative proximity cannot be excluded.

Features of Imprinting

An imprint, which can be considered a physical differential modification of a particular chromatin region, has been suggested from previous data to have the following characteristics: once introduced, it remains heritable through subsequent rounds of DNA replication; it affects the expression of the imprinted gene; and it is reversibly switched following passage through the germline of opposite sex (Sapienza et al., 1989). Since the imprint can be erased, it is obligatory to postulate that marking is conferred by an activity (or factor), the product of an imprinting gene, that is expressed exclusively in one of the germ lines (Sapienza, 1989).

From the evidence that we have presented, we deduce that the putative imprint associated with the IGF-II locus exhibits all of the ascribed features, with the notable exception that both alleles are expressed in the plexus, in contrast to the sole expression of the paternal allele in other tissues. If the IGF-II gene is typical of an imprinted locus, this exception could mean that an imprint can be erased in a tissue-specific manner. Alternatively, the imprint may still be maintained, but not recognized. In this regard an important question is whether imprinting is positive or negative. Although it can be postulated that opposite imprinting signals are conferred on both the paternal and maternal chromosomes, a simpler assumption is that only one chromosomal homolog is modified, while the other remains at a default state.

If imprinting is positive, the imprint is present on the paternal IGF-II gene and is recognized, directly or indirectly, by a tissue-specific "activator" modulating transcription, whereas the unmarked maternal allele remains silent. We do not favor this model because it is unlikely that a germline-specific imprinting activity could be expressed in the plexus to mark the maternal allele positively. Thus, negative imprinting seems more compatible with our data. In this scenario the paternal gene is expressed in embryos by default via the action of tissue-specific *trans*-acting factors. The latter are not expressed in adult tissues, with the exception of the plexus. The imprint then is associated with the maternal allele and is recognized by a "repressor" whose expression is extinguished from the plexus cells some time before e16.5, allowing maternal gene transcription to occur.

Regardless of the actual mode of action of the imprinting process, an important conclusion from our observation of expression of the maternal IGF-II allele in the plexus is that developmentally regulated DNA rearrangements occurring in the germline as a means of imprinting are highly unlikely.

Imprinting Mechanisms

The strain of mice carrying a tagged null mutation at the

endogenous IGF-II gene locus is a model system that can be used to investigate the molecular nature of the imprinting mechanism operating in one of the two classes of imprinted loci, whereby the paternal allele is active and the maternal allele silent.

Estimates from the genetic experiments involving chromosomal rearrangements suggest that only a minority of mouse genes (about 10%) are potentially subject to imprinting (see Cattanach, 1986; Surani et al., 1990). Thus, regardless of the particular details of any model, it is obligatory to postulate that the imprinting activity recognizes a specific DNA sequence, an *imprinting box*. The imprint then could be either a covalent DNA modification in the context of this box, introduced by the imprinting activity, or a complex between the box and the imprinting factor. The latter possibility seems less likely because only complicated models could account for the differential maintenance of such a complex during DNA replication. Thus, 5-methylcytosine, which is the only modified base identified to date in mouse DNA (reviewed by Adams and Burdon, 1985), could be involved in imprinting. If this is correct, the imprinting activity could be a germline-specific "imprinting methylase." The appeal of this model is that the heritability and erasure of an imprint can readily be explained by known mechanisms.

The specific C-methylation of the imprinting box that we propose should be considered as distinct from the general methylation patterns associated with particular gene regions or embryonic stages (reviewed by Cedar, 1988; Monk, 1990; Cedar and Razin, 1990). Previously, a connection between methylation and imprinting was made on the basis of the observation that C-methylation patterns of particular transgenes can depend on their parental origin (reviewed by Surani et al., 1988). However, "methylation imprinting" did not correlate with transgene expression, with one exception (Swain et al., 1987). In this case the paternal transgene was undermethylated and expressed exclusively in the myocardium, while the maternal transgene, which was hypermethylated and not expressible, could switch when transmitted through the male germline. However, when the genomic target site of this transgene was cloned, it did not exhibit methylation imprinting behavior (R. Chaillet and P. Leder, personal communication).

Imprinting of the IGF-II Receptor Gene

Recently, study of the *Tme* deletion (T-associated maternal effect) has provided another specific example of an imprinted gene (Barlow et al., 1991). Embryos that inherit the deletion maternally die at day 15 of gestation, but if the deletion is paternally transmitted, the offspring are viable (Johnson, 1974, 1975; McGrath and Solter, 1984b). Four known genes have been mapped in the region of the *Tme* deletion (encompassing approximately 0.9 Mb), including the gene encoding the IGF-II receptor (*Igf2r*). It was shown that, in contrast to the other three loci, the *Igf2r* locus is transcriptionally active only when maternally derived (Barlow et al., 1991). Thus it is very likely that it is imprinted in a reciprocal manner to the IGF-II gene. However, other features of imprinting have yet to be demonstrated.

Despite the coincidental discoveries demonstrating im-

printing in both the IGF-II and IGF-II receptor genes, it is unlikely that the two phenomena are related. The mammalian IGF-II receptor is a bifunctional molecule that also serves as the cation-independent mannose-6-phosphate receptor (CI-MPR) involved in lysosomal enzyme targeting (reviewed by Roth, 1988; Kornfeld and Mellman, 1989). In contrast, the *Xenopus* and the chicken CI-MPRs lack a high affinity binding site for IGF-II (Canfield and Kornfeld, 1989; Clairmont and Czech, 1989). Thus it is likely that the growth factor exerts its function via the IGF-I receptor, to which it binds, although with lower affinity than IGF-I (Steele-Perkins et al., 1988). Currently there is no compelling evidence to suggest that the mammalian IGF-II/CI-MPR participates in IGF-II signal transduction. The lack of a relationship between the developmental role of IGF-II and the IGF-II/CI-MPR is also indicated from the disparity in mutant phenotypes. Targeted mutagenesis of the IGF-II gene results in small but viable mice, whereas the lethality of the embryos with the *Tme* deletion is not associated with growth deficiency (Barlow et al., 1991). If the ligand and the receptor were related in terms of developmental function, the phenotypes would have been similar, regardless of whether or not the *Igf2r* gene is proven to be the *Tme* gene.

The identification of specific imprinted loci is expected to accelerate studies on the molecular mechanisms underlying the imprinting process, although elucidation of the biological significance of this phenomenon may continue to remain elusive. At present, the differential expression of imprinted genes can be viewed as a mechanism of functional gene dosage adjustment. This, however, does not explain the selective advantage that led to the evolutionary emergence of imprinting in mammals, which seems to be unique among the vertebrate phyla.

Experimental Procedures

Mice

The genetic analysis was performed by initiating colonies from two complete and one partial germline male chimeras, described previously (animals A, B, and C; Table 2b in DeChiara et al., 1990). These founders, and their heterozygous progeny of subsequent generations, were mated with wild-type mice of strains 129/Sv/Ev (from our own colony); MF1 and BALB/c (from Harlan Sprague Dawley, Inc., Indianapolis, Indiana); and C57BL/6J and (C57BL/6J × DBA/2J)F1 (from the Jackson Laboratories, Bar Harbor, Maine). Consistent results were obtained, regardless of strain.

DNA and RNA Analyses

For genotyping, high molecular weight DNA was prepared from tail biopsies or from embryonic tissue and analyzed by Southern blotting as described (DeChiara et al., 1990). Total cell RNA was prepared from embryos or placentas by the acid guanidine thiocyanate method (Chomczynski and Sacchi, 1987), and from brain by the guanidine thiocyanate-CsCl procedure (Chirgwin et al., 1979). For nuclease protection analysis, we used uniformly labeled E2 (147 nucleotide), *neo* (246 nucleotide), and actin (380 nucleotide) riboprobes that were synthesized in vitro from linearized DNA templates, using T7 RNA polymerase (Pharmacia) and [³²P]UTP (800 Ci/mmol; Du Pont). The sizes of the protected RNA fragments were 135, 182 and 350 nucleotides, respectively. The following DNA templates were used: A BamHI-SacII DNA fragment (135 bp) representing most of the exon E2 sequence was isolated from the rat IGF-II cDNA clone 27 (Soares et al., 1986) and subcloned into pGEM-1 (Promega). The template for the *neo* riboprobe (probe 2; Figure 1) was a 182 bp EcoRI fragment from pMC1PolA (coor-

dinates 447–629; Stratagene), representing regulatory sequences of the *neo* cassette, that was subcloned into the EcoRI site of pBluescript SK+ (Stratagene). The mouse cytoplasmic actin probe was prepared from pGEMmAct350 (provided by Dr. Elizabeth Bikoff) that contains a 350 bp PstI-HindIII cDNA fragment, subcloned into the corresponding sites of pGEM-4 (Promega).

RNAase protection analysis was performed as described (Sambrook et al., 1989), with minor modifications. The RNA (brain, 25 µg per assay; embryo or placenta, 20 µg per assay) was hybridized with combinations of probes (using approximately 5×10^4 cpm of the actin probe and 1.5×10^5 cpm of either the E2 or *neo* probe per assay; see Figure 4) in a volume of 20 µl containing 80% deionized formamide (Aldrich), 40 mM PIPES (pH 6.4), 400 mM NaCl, and 1 mM EDTA. The samples were denatured at 68°C for 10 min and then incubated at 45°C for 12–16 hr. Unhybridized probes were digested at 30°C for 2 hr with 50 µg/ml RNAase A and 2 µg/ml RNAase T1 (Sigma) in a volume of 360 µl. The samples were then treated with 1 mg/ml proteinase K at 37°C for 15 min, phenol extracted, ethanol precipitated, and electrophoresed in denaturing 5% polyacrylamide-urea gels. Autoradiography was for 16–72 hr with an intensifying screen.

In Situ Hybridization

For staging of embryos, we considered the morning of observing a vaginal plug as e0.5. Embryos were sectioned and prepared for in situ hybridization as described (Wilkinson et al., 1987). The antisense E2 riboprobe described above was labeled with [³⁵S]UTP (1000–1500 Ci/mmol; Du Pont). An E2 sense (control) probe was synthesized from the same template linearized with PvuII, using SP6 RNA polymerase (Promega). The *neo* probe (probe 3; Figure 1) was prepared from a template consisting of a 962 bp EcoRI-BamHI fragment of pMC1PolA (coordinates 629–1591) that was subcloned into pBSK+. Following linearization with BamHI, an antisense probe (1015 nucleotides) was synthesized using T3 RNA polymerase (Promega). Hybridization, exposure to photoemulsion, counterstaining, and microphotography were as described (Lugo et al., 1989).

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